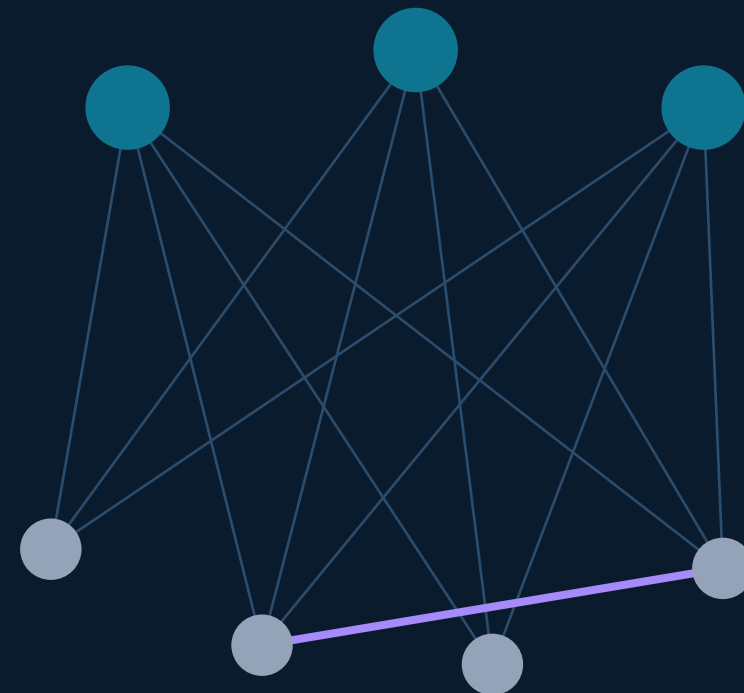


NETWORK AS CODE · DMVPN PHASE 3

C8000v DMVPN Portal

One place to see, change, prove and share a multi-hub DMVPN service:
for the operators who run it and the customers who use it.



Three Catalyst 8000v hubs · C8000v and VyOS customers · two providers · Nautobot · Terraform (NAC) · VictoriaMetrics

Version 0.21.0 · September 2026

Running a DMVPN service by hand does not scale

A single customer touches three hubs, one or two providers, two router operating systems and a source of truth. Every question and every change means logging in to several boxes.

Scattered state

Is every spoke registered with every hub? Are shortcuts forming? Health lives in a dozen CLI sessions on IOS XE and VyOS.

Slow, risky changes

Adding, changing or removing a customer means consistent edits on hubs, spoke, provider and Nautobot. It is easy to miss one.

Drift

Hand edits leave routers differing from the intended design, and nobody notices until something breaks.

No change control

Who changed what, who approved it, and what exactly changed on each router? Usually nobody can say.

Unproven resilience

Dual hubs and dual providers look good on paper, but how long does failover really take?

Blind customers

Customers cannot see their own service, test it, or ask for changes. SLA reports are assembled by hand.

Static secrets

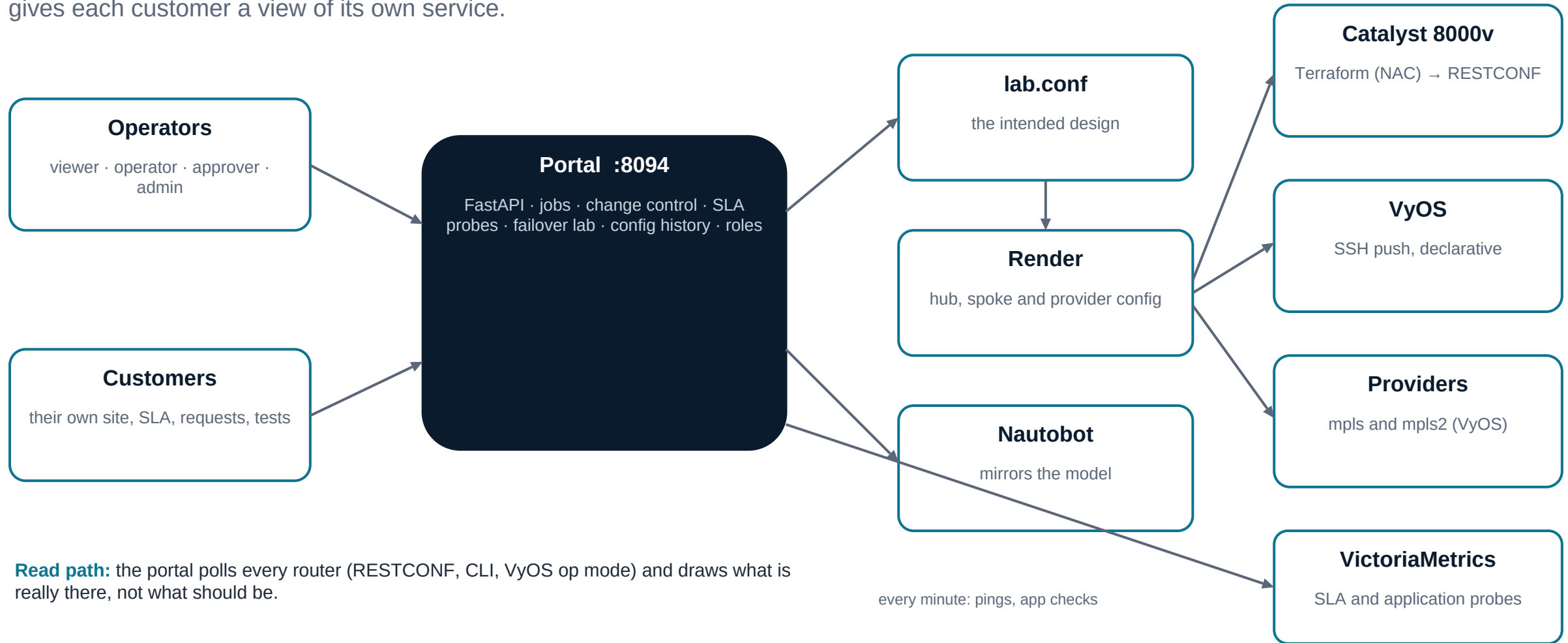
The DMVPN pre-shared key is set once and rarely rotated, because rotating it by hand risks an outage.

The result

Slow onboarding, outages from avoidable mistakes, and no evidence for customers or auditors.

What the portal does

One web portal on top of a single source of truth (**lab.conf**). It reads every router live, turns every change into a controlled, verified job, and gives each customer a view of its own service.



Read path: the portal polls every router (RESTCONF, CLI, VyOS op mode) and draws what is really there, not what should be.

Capabilities at a glance

1 See

Live cloud table and KPIs, a topology map drawn from live state, per-customer details, path trace and 'watch the shortcut form'.

2 Change

Add, modify and remove customers, deploy the model, fix drift, back up and restore, choose the preferred hub or add a second provider.

3 Control

Change requests with four-eyes approval, change windows, emergency changes, and maintenance mode that mutes alerts.

4 Audit

Every job snapshots every router before and after it runs, records who asked and who approved, and shows a diff per router.

5 Prove

SLA per customer (availability, latency, loss), application checks, simulated failures and measured failover times.

6 Secure

Sign-in with roles, admin-only pre-shared key rotation with a random key, and secrets masked in every view and log.

7 Customers

A read-only customer portal: its own site and applications, SLA and PDF report, change requests and self-service tests.

8 Automate

A REST API for everything (Swagger), Prometheus metrics, progress and ETA for each job, and a 73-test Robot suite after each change.



A tour of the portal

Numbered callouts on each screenshot match the explanations beside it.

Everyone signs in, and the role decides what they see

The screenshot displays the C8000v DMVPN Portal interface. At the top, there's a navigation bar with links like 'The cloud', 'Network map', 'SLA', 'Resilience', 'Provision', 'Jobs', 'Lab Tools', and 'API'. Below this, the main content area is divided into sections: 'The cloud' (with a 'Refresh' button), 'The provider', 'Health', and 'Configuration drift'. A sign-in modal is overlaid on the 'The cloud' section. The modal has a title 'Sign in', fields for 'Username' and 'Password', and a 'Sign in' button. Below the button, there's a note: 'Staff accounts see the whole lab by role; a customer account sees its own service. The lab-default accounts are in webapp/users.seed.json — change their passwords.' A red circle with the number '1' is placed over the sign-in modal.

1

Sign in

Staff roles are viewer, operator, approver and admin. A customer account sees only its own service. Sessions are signed cookies, passwords are PBKDF2 hashes.

Why it matters

Read-only users cannot start jobs, nobody approves their own change, and customers never see another customer.

- 1 KPIs**
Hubs, customers, registrations, shortcuts and BGP at a glance.
- 2 Every router**
NHRP registrations, IPsec, overlay BGP, the provider and the preferred hub, read from the routers themselves.
- 3 Both providers**
Access links and eBGP on mpls and mpls2.
- 4 Who you are**
The account, its role and the portal version, always visible.

One screen answers 'is every spoke registered with every hub?' with no CLI logins.

SEE

Health and configuration drift

Prairie Grain Logistics

cust4 East

customer

172.28.0.14

100.70.14.2

3 / 3 hubs

cloud 2: 3 / 3

8

6 / 6

both established

none

192.168.64.0/24

host-cust4

up

Redwood Coast Brewing Co.

cust5 Central

customer

172.28.0.15

100.70.15.2

3 / 3 hubs

5

3 / 3

established

none

192.168.65.0/24

host-cust5

up

Summit Ridge Property Management

cust6 West

customer

172.28.0.16

100.70.16.2

3 / 3 hubs

5

3 / 3

established

none

192.168.66.0/24

host-cust6

up

Gulfstream Marine Services

A customer registers with every hub, so NHRP should read 3 / 3. Shortcuts are customer-to-customer tunnels NHRP has resolved (phase 3); they come and go with traffic.

The provider

a VyOS router standing in for the MPLS network: it carries the sites' WAN addresses and nothing else

Router	AS	Accepts	Interfaces up	eBGP customers	Attached
mpls primary	AS 65000	100.70.0.0/16	9 up	8 / 8	hub-east (eth1), hub-central (eth2), hub-west (eth3), cust1 (eth4), cust2 (eth5), cust5 (eth6), cust4 (eth7), cust6 (eth8)
mpls2 second: the backup cloud	AS 65010	100.71.0.0/16	6 up	5 / 5	hub-east (eth1), hub-central (eth2), hub-west (eth3), cust4 (eth4), cust1 (eth5)

1

Health

Everything the model expects is true

2

Configuration drift

checked 9/27/2026, 5:38:23 PM · took 18s · every 6 h while the lab runs

Check now

Fix drift

no drift: every router runs what the model says

Nautobot == lab.conf

Router	Verdict	Differences (missing: the model has it, the router does not · extra: the router has it, the model does not · changed: Terraform would update it)
cust1 C8000v customer	matches the model	
cust2 C8000v customer	matches the model	
cust4 VyOS customer	matches the model	
cust5 VyOS customer	matches the model	
cust6 VyOS customer	matches the model	
hub-central C8000v hub	matches the model	
hub-east C8000v hub	matches the model	
hub-west C8000v hub	matches the model	
mpls VyOS provider	matches the model	
mpls2 VyOS provider	matches the model	

- 1

Health

Everything the model expects to be up is checked, and anything missing is named.
- 2

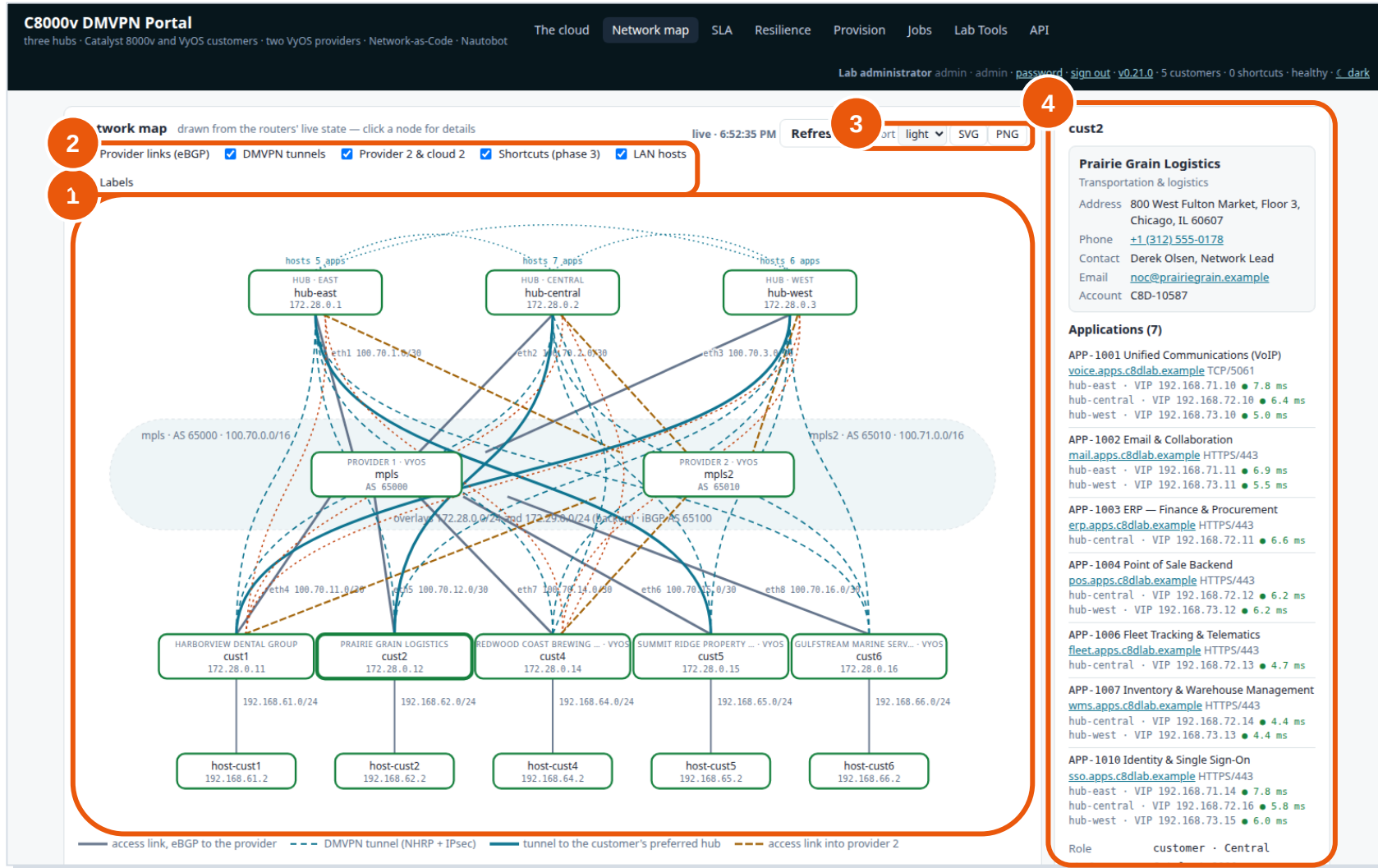
Drift

Each router is compared with what lab.conf renders. 'Fix drift' puts it back as a controlled job.

Why it matters

Hand edits no longer go unnoticed: drift is visible per router, and fixing it is one approved job.

A topology map drawn from live state



1

The map

Hubs, providers, customers and LAN hosts. Tunnel colour shows health, and the preferred hub is drawn bold.

2

Layers

Turn provider 2 and cloud 2, shortcuts, hosts and labels on or off.

3

Export

SVG or PNG, light or dark, exactly as drawn.

4

Details

Click a node to see the customer company, its applications (checked every minute) and the router's live state.

Why it matters

Everyone gets the same live picture, from operations to account managers and customers.

Path trace: watch the phase 3 shortcut form

1

hosts 5 apps: HUB - EAST (172.28.0.1), HUB - CENTRAL (172.28.0.2), HUB - WEST (172.28.0.3)

hosts 7 apps: HUB - CENTRAL (172.28.0.2)

hosts 6 apps: HUB - WEST (172.28.0.3)

mpls AS 65000: 100.70.0.0/16

PROVIDER 1 - VYOS mpls AS 65000

PROVIDER 2 - VYOS mpls2 AS 65010

mpls2 AS 65010: 100.71.0.0/16

HARBORVIEW DENTAL GROUP cust1 (172.28.0.11)

PRAIRIE GRAIN LOGISTICS cust2 (172.28.0.12)

REDWOOD COAST BREWING ... - VYOS cust4 (172.28.0.14)

SUMMIT RIDGE PROPERTY ... - VYOS cust5 (172.28.0.15)

GULFSTREAM MARINE SERV... - VYOS cust6 (172.28.0.16)

host-cust1 (192.168.61.2)

host-cust2 (192.168.62.2)

host-cust4 (192.168.64.2)

host-cust5 (192.168.65.2)

host-cust6 (192.168.66.2)

Legend:

- access link, eBGP to the provider
- DMVPN tunnel (NHRP + IPsec)
- tunnel to the customer's preferred hub
- access link into provider 2
- cloud 2 tunnel (backup)
- live customer-to-customer shortcut
- down
- traced path through a hub
- configuration drift
- traced path over the shortcut
- outline: healthy · partial · down / unreachable

2

cust4

Address 300 Harbor Way, Eureka, CA 95501

Phone +1 (707) 555-0168

Contact Casey Lindqvist, Operations IT

Email it@redwoodcoast.example

Account C8D-10692

Role host

Platform Alpine Linux

Status healthy

VM c8d-host-cust4

Management 10.5.0.44

LAN address 192.168.64.2/24

Gateway 192.168.64.1 (cust4)

Login lab / lab

show configuration

ping all hosts (4)

connect to another host (traceroute, drawn on the map)

host-cust5 (cust5) trace path

watch the shortcut form clear path

3

host-cust4 → host-cust5: watching the shortcut form 6:53:00 PM direct (shortcut)

```
# clearing the shortcut between cust4 and cust5...
# cust4: clear ip nhrp cache (VyOS: every dynamic entry)
# cust5: clear ip nhrp cache (VyOS: every dynamic entry)
```

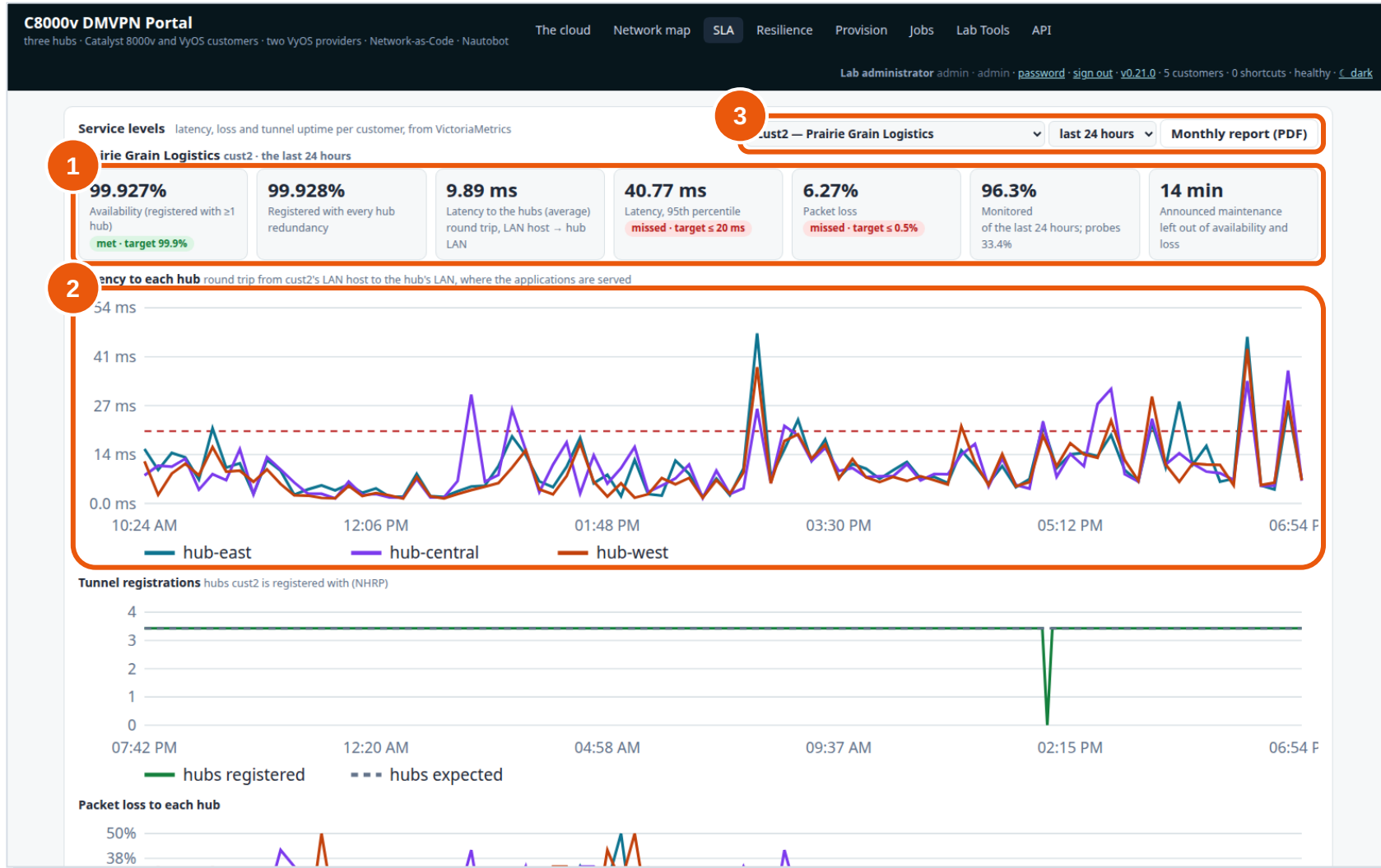
Copy Download Close

- 1 First via a hub, then direct**
The first packets go through a hub. NHRP then builds a spoke-to-spoke shortcut, drawn in purple.
- 2 Details and actions**
Trace, watch the shortcut form, ping every host, and view the config history per router.
- 3 Every trace, hop by hop**
The raw output is kept so you can copy or download it.

Why it matters

It shows that DMVPN phase 3 works end to end, and makes it easy to explain.

SLA per customer, from VictoriaMetrics



1 Against targets
Availability, latency, loss and tunnel uptime, each compared with its SLA target.

2 Latency to each hub
Probed every minute from the customer's own LAN host.

3 Window and report
24 hours, 30 days or this month, with a monthly PDF report for the customer.

Why it matters

SLA reporting is automatic and based on measurements the customer can check.

Simulate failures on purpose

C8000v DMVPN Portal

three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloudNetwork mapSLAResilienceProvisionJobsLab ToolsAPI

Lab administratoradmin · admin · password · sign out · v0.21.0 · 5 customers · 0 shortcuts · healthy · [C_dark](#)

1

What happens: the hub's VM is frozen: no packets, no BGP, no NHRP

What the design should do: customers keep two hubs; routes through it move to the others when BGP gives up on it (hold 9 s); live shortcuts do not need the hub and stay up

Fail it nowMeasure failover

2

Where

hub-east

Hold (failover test)

60

In now

a failure left in is put back after 30 min

nothing is failed right now

Restore all

3

Failover measurements

every LAN host pings every other host and every hub's LAN five times a second while the fault is in

When	Failure	Held	Flows	Worst failover	Back to healthy
Sep 27, 01:49:13 PM	Hub fails on hub-east	45 s	29 unaffected 1 failed over 5 cut off	0.6 s	56.1 s
Sep 27, 01:44:24 PM	Provider fails on mpls	60 s	8 failed over 27 cut off	48.6 s	168.4 s
Sep 27, 01:34:12 PM	Provider fails on mpls	60 s	4 failed over 31 cut off	6.8 s	165.8 s

Earlier failures

Failure	Where	In	Out	How it ended
Customer circuit cut	cust1:wan2	Sep 27, 05:38:36 PM	Sep 27, 05:38:49 PM (13 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 03:53:23 PM	Sep 27, 03:53:34 PM (11 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 03:36:27 PM	Sep 27, 03:36:38 PM (10 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 03:26:14 PM	Sep 27, 03:26:26 PM (11 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 03:21:17 PM	Sep 27, 03:21:27 PM (10 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 02:45:05 PM	Sep 27, 02:45:14 PM (9 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 02:22:14 PM	Sep 27, 02:22:24 PM (10 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 02:12:08 PM	Sep 27, 02:12:17 PM (9 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 02:04:58 PM	Sep 27, 02:05:09 PM (11 s)	restored

- 1

Choose a failure

A hub, a provider, or a customer circuit. The page says what should happen before you press anything.
- 2

Measure failover

Every LAN host pings every other host five times a second while the fault is in.
- 3

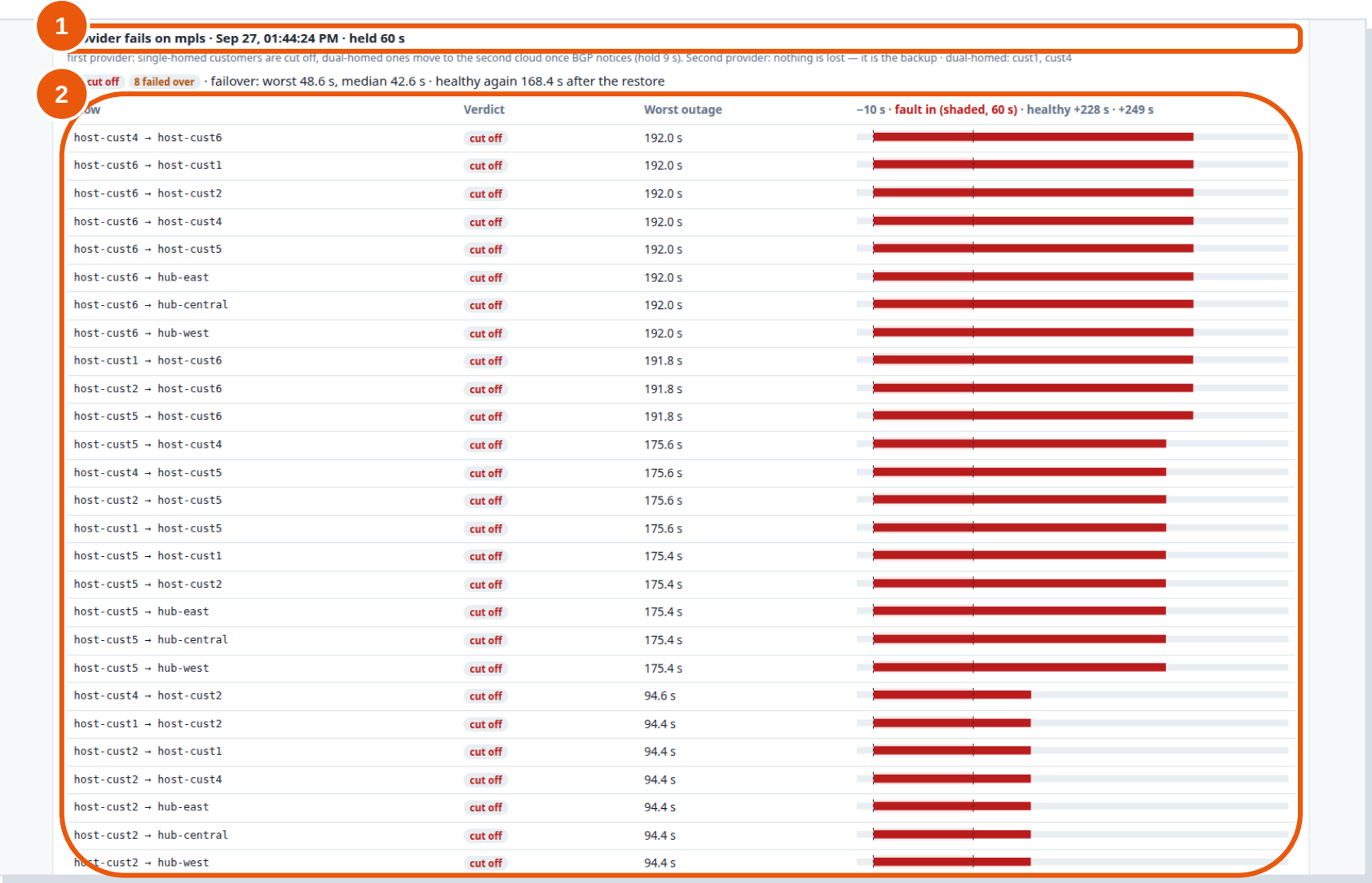
Every measurement

Flows unaffected, failed over or cut off, the worst failover and the time back to healthy.

Why it matters

Resilience becomes a number rather than a claim, for example 0.6 s worst case for a hub failure.

One failover experiment, flow by flow



- 1

The experiment
What failed, for how long, and the headline figures.
- 2

Each flow
The verdict and an outage timeline for every host pair, so you can see which paths moved and which were cut.

Why it matters
You can see exactly which customers a failure affects, and for how long.

- Tasks**
Add, modify and remove customers, deploy, check for and fix drift, back up, restore, dry run and tests.
- Pre-shared key**
Admins rotate the DMVPN key. The system picks a random 40-character key, and only its fingerprint is ever shown.

Operators pick what they want done, and the portal carries it out consistently on every router.

Modify a customer, with the plan shown first

C8000v DMVPN Portal
three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloud Network map SLA Resilience Provision Jobs Lab Tools API

What do you want to do? every change

Add a customer
A new site: a C8000v and its LAN host, a link into the provider, Tunnel0 registered with all three hubs. No hub changes. about 15 minutes (C8000v first boot)

Check for drift
Compare every router's running configuration with the model: Nautobot vs lab.conf, terraform plan for the C8000vs, the rendered set lines for VyOS. Changes nothing. read only · a minute or two

Run the tests
The Robot Framework suites against the live lab, with the evidence kept. about 10 minutes

DMVPN pre-shared key the IKEv2 key

Key	Set	Used by
secrets/dmvpn...	Sep 27, 05:20:53 PM	hub-east, hub-cent

A rotation picks a random 40-character key, appears here, in the logs or in git: it is in secret

Modify a customer

Customer: cust2 — Prairie Grain Logistics (Catalyst 8000v, C8000v)

1 Router type: Cisco Catalyst 8000v

2 Preferred hub: hub-central (Central)

3 Second provider: dual-homed

Region: Central Site LAN: 192.168.62.0/24

Management address: 10.5.0.22 Index (WAN /30, tunnel, router-id): 12 Provider port: mpls eth5 LAN host: host-cust2 Host management: 10.5.0.42

The customer company

Company: Prairie Grain Logistics Industry: Transportation & logistics Contact: Derek Olsen, Network Lead

Address: 800 West Fulton Market, Floor 3, Chicago, IL 60607 Phone: +1 (312) 555-0178 Email: noc@prairiegrain.example

Applications (what the customer subscribes to)

☒ APP-1001 Unified Communications (VoIP)	☒ APP-1002 Email & Collaboration	☒ APP-1003 ERP — Finance & Procurement
☒ APP-1004 Point of Sale Backend	☐ APP-1005 Electronic Health Records	☒ APP-1006 Fleet Tracking & Telematics
☒ APP-1007 Inventory & Warehouse Management	☐ APP-1008 Secure File Transfer	☐ APP-1009 Video Surveillance (NVR)
☒ APP-1010 Identity & Single Sign-On		

4 Second provider: single-homed → dual-homed (mpls2 eth6, 100.71.12.0/30)

The job

- cust2 is restarted with its port 4 wired into mpls2 eth6 (its configuration is saved first; a C8000v is back in ~5 min, a VyOS in ~1); then Tunnel1 registers with every hub in the second cloud
- Network-as-Code applied (C8000v) and the VyOS routers re-pushed; the hubs do not change
- Nautobot re-modelled: the customer is taken out and seeded again

Re-wiring port 4 restarts the router: the site is offline for a few minutes.

☒ run the suites at the end

Apply the changes **Re-check** **Cancel**

Dry run
terraform plan against the routers, and Nautobot's rendering compared with lab.conf's. Changes nothing. read only

Rotate the pre-shared key
A new random IKE pre-shared key on every router (it never leaves secrets/ on the lab host), then every router's IKE sessions set up again and checked. admin · about 7 minutes

Rotate the pre-shared key

as made with the new key. The key never

- Router type**
Move a customer between C8000v and VyOS.
- Preferred hub**
Pick the hub its traffic should prefer (local-preference 200).
- Second provider**
Dual-home the customer onto mpls2 and cloud 2.
- The plan**
Exactly what will change on which routers, before anything runs.

Why it matters

No surprises: the impact is visible and reviewed before a change request is filed.

Change requests: four eyes and change windows

C8000v DMVPN Portal
three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloud Network map SLA Resilience Provision **Jobs** Lab Tools API

Lab administrator admin · admin · password · sign out · v0.21.0 · 5 customers · 1 shortcuts · healthy · [C_dark](#)

Change requests approval for: remove, modify, restore, deploy, fixdrift, failover, fault, rotatepsk · four eyes · windows off [Change policy...](#)

1 accounts still on their lab-default password: admin, approver, operator, viewer — change them in Lab Tools → Users.

Change	What	Asked by	Status
CR-0023 Sep 27, 05:39:15 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected
CR-0022 Sep 27, 05:38:29 PM	Simulate a failure simulate: Customer circuit cut on cust1:wan2 why: suite 07	operator approved by approver: suite 07	started
CR-0021 Sep 27, 05:20:49 PM	Rotate the pre-shared key rotate the IKE pre-shared key on every router why: first rotation off the public lab default	admin approved by approver: rotate	done job
CR-0020 Sep 27, 03:53:59 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected
CR-0019 Sep 27, 03:53:20 PM	Simulate a failure simulate: Customer circuit cut on cust1:wan2 why: suite 07	operator approved by approver: suite 07	started
CR-0018 Sep 27, 03:36:59 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected
CR-0017 Sep 27, 03:36:24 PM	Simulate a failure simulate: Customer circuit cut on cust1:wan2 why: suite 07	operator approved by approver: suite 07	started
CR-0016 Sep 27, 03:26:55 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected
CR-0015 Sep 27, 03:26:11 PM	Simulate a failure simulate: Customer circuit cut on cust1:wan2 why: suite 07	operator approved by approver: suite 07	started
CR-0014 Sep 27, 03:21:54 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected

1

Change requests

Each request shows who asked, what for, who approved or rejected it, and its status, including requests from customers.

Why it matters

Disruptive work cannot start without a second person, and it runs only inside the agreed windows.

A job: who, what, and the proof

The screenshot displays a pipeline audit interface with four numbered sections:

- Who:** A green bar at the top indicates the job was started by 'admin' and approved by 'approver' with a 'success' status. A progress bar shows '100% - 9 / 9 steps' and 'took 5m 28s'.
- Progress:** A list of 9 steps, each marked as 'success':
 - Snapshot every router's configuration (before) every router saved
 - Choose a new random pre-shared key (kept in secrets/, never in git or the log) new key ...552085ff (the key itself is in secrets/dmvpn_psk only)
 - Render the configuration from lab.conf configuration re-rendered from lab.conf
 - Network-as-Code: terraform apply (resources on the new router only) applied and saved
 - The provider: address the new access link; the customer arrives on its listen range provider re-pushed (idempotent; a changed port is reconciled)
 - Re-authenticate every IKE session, one router at a time every router's IKE sessions set up again
 - Verify every IKE session: how it authenticated, and that it is new 54 IKE SAs on 8 routers, all authenticated with the pre-shared key, all made after the rotation
 - Verify: registered with every hub, IPsec and BGP up, hosts reach each other, Nautobot == lab.conf the cloud is healthy; every host reaches every other; Nautobot == lab.conf
 - Snapshot every router's configuration (after) and compare cust1 +1/-1; cust2 +1/-1; cust4 +1/-1; cust5 +1/-1; cust6 +1/-1; hub-central +1/-1; hub-east +1/-1; hub-west +1/-1
- What changed:** A section titled 'configuration changes:' with a list of routers: cust1 +1/-1, cust2 +1/-1, cust4 +1/-1, cust5 +1/-1, cust6 +1/-1, hub-central +1/-1, hub-east +1/-1, hub-west +1/-1. A 'show the differences' button is present.
- Every step:** Two expanded diff views for 'cust1 before' and 'cust2 before' configurations. The 'after' state shows the pre-shared key being updated from 'd76a' to '552085ff'.

- Who**
Who started the job and who approved it.
- Progress**
Live progress and the time it took.
- What changed**
The before-and-after diff per router. Secrets are masked but their change is still visible.
- Every step**
Each step's result, for example '54 IKE SAs on 8 routers, all made after the rotation'.

Why it matters

Every change leaves evidence: the steps, the approvals and the exact configuration difference.

Lab tools and accounts

C8000v DMVPN Portal

three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloudNetwork mapSLAResilienceProvisionJobsLab ToolsAPI

Lab administratoradmin · admin · password · sign out · v0.21.0 · 5 customers · 1 shortcuts · healthy · [C_dark](#)

1

Users

who may sign in, and what each role may do — viewer reads, operator changes, approver approves, admin all, customer its own service

Account	Name	Roles	Customer	State	
admin	Lab administrator	admin		active lab-default password	set passworddisable
approver	Change approver	approver		active lab-default password	set passworddisabledelete
operator	Network operator	operator		active lab-default password	set passworddisabledelete
prairie	Derek Olsen (Prairie Grain Logistics)	customer	cust2	active	set passworddisabledelete
viewer	Read-only viewer	viewer		active lab-default password	set passworddisabledelete

Username

Full name

Role

Password (8+)

viewer

Add the account

How to get in

Lab host: ssh dcantor@192.168.50.231 then cd /home/dcantor/c8000v-dmvpn-lab management network 10.5.0.0/24 (c8d-oob, host 10.5.0.1)

The management addresses (10.5.0.0/24) and the consoles live on the lab host (bridge c8d-oob, host 10.5.0.1): log in to the lab host first, then ssh from there or run ./lab.sh in /home/dcantor/c8000v-dmvpn-lab.

2

Tools

everything the lab uses, with the login each one takes

LAB

Provisioning portal

the cloud, the Network map, provisioning jobs; customers sign in to their own view

login: an account: admin / operator / approver / viewer, lab defaults in webapp/users.seed.json

http://192.168.50.231:8094

Portal API (Swagger)

every endpoint the portal offers; /metrics and /api/sd for Prometheus stay open

login: the portal's session (sign in first)

http://192.168.50.231:8094/docs

Lab hub

every lab on this host, VM power, host CPU / memory

login: none

http://192.168.50.231:8088

GitHub repository

the code: lab.sh, the renderer, NAC, tests, Nautobot, the portal

login: public

https://github.com/dcantor/c8000v-dmvpn-lab

SOURCE OF TRUTH

Nautobot

the shared NMS (cat9000v lab); API token: ./lab.sh nautobot token

Nautobot: this lab's devices

every VM of the lab, under its c8d- name

Nautobot: customer companies

a tenant per customer, with its details and subscriptions

Nautobot: applications (Virtual Servers)

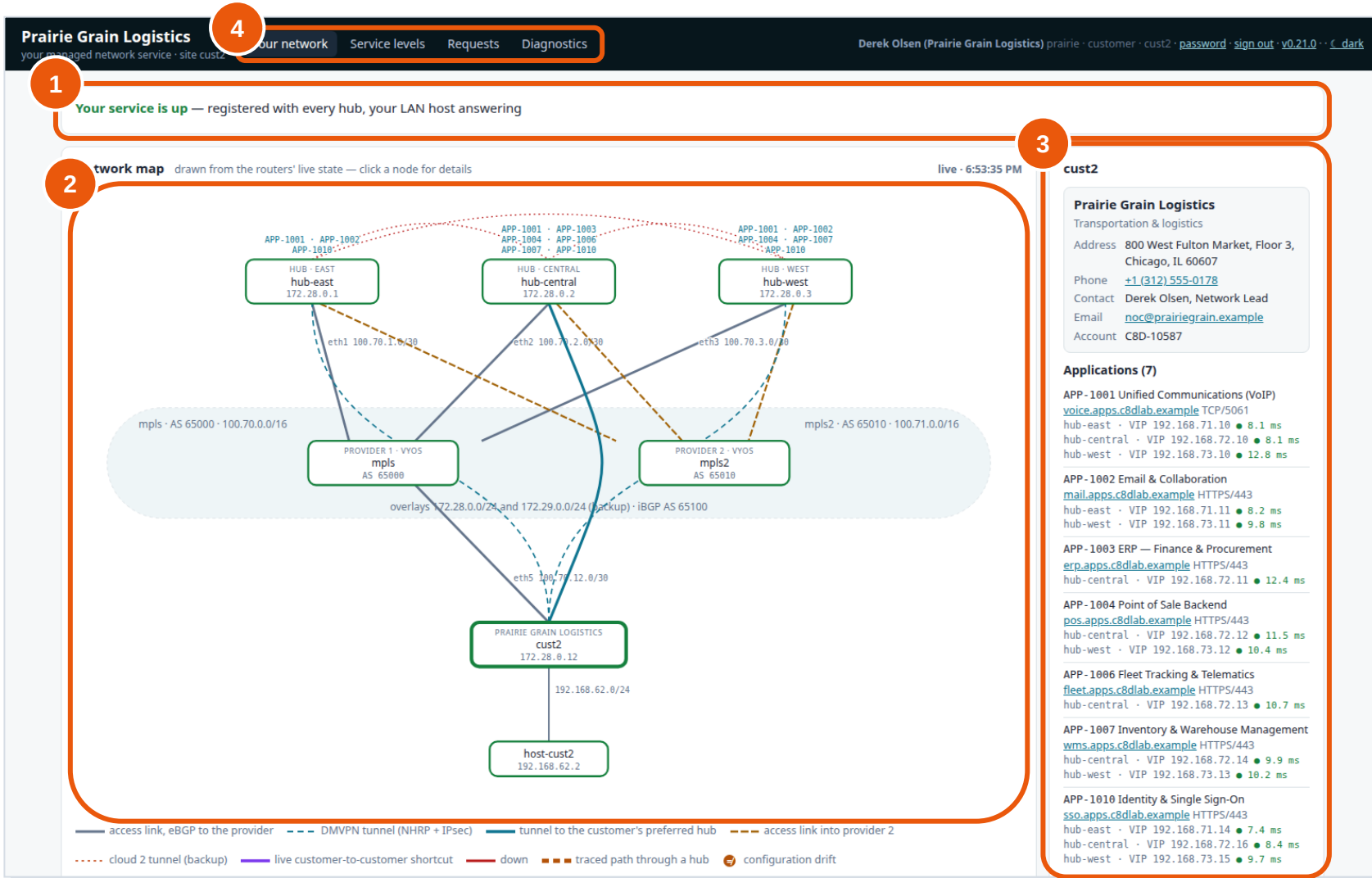
one per application per hosting hub, with its VIP

1 **Accounts and roles**
Admins add, disable and delete accounts and set passwords. Accounts still on their lab-default password are flagged.

2 **Every tool**
Nautobot, the API, the lab hub and the repository, each with how to sign in.

Why it matters
Access is managed in one place, with visible warnings when the defaults have not been changed.

The customer's own view



- 1 Status**
Service status, incidents and planned maintenance for this customer.
- 2 Its own site**
The map, cut down to the customer's own site and the hubs it uses.
- 3 Its applications**
Its company details and live application status.
- 4 Its own pages**
Network, service levels, requests and diagnostics. Nothing else is reachable.

Why it matters

Customers can answer their own questions, which means fewer tickets and more trust.

Customer self-service requests

Prairie Grain Logistics

your managed network service · site cust2

Your network

Service levels

Requests

Diagnostics

Derek Olsen (Prairie Grain Logistics)

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1

Ask for a change

your request goes to our operators as a change request; you see here when it is approved and done

Applications (what you subscribe to)

☒ APP-1001 Unified Communications (VoIP)

☒ APP-1002 Email & Collaboration

☒ APP-1003 ERP — Finance & Procurement

☒ APP-1004 Point of Sale Backend

☐ APP-1005 Electronic Health Records

☒ APP-1006 Fleet Tracking & Telematics

☒ APP-1007 Inventory & Warehouse Management

☐ APP-1008 Secure File Transfer

☐ APP-1009 Video Surveillance (NVR)

☒ APP-1010 Identity & Single Sign-On

Preferred hub

hub-central (Central)

Second provider

☐ a second, backup circuit (dual-homed)

Why (optional)

Send the request

2

My requests

Request	What	Asked	Status
CR-0012	modify cust2: preferred hub none → hub-central; applications APP-1001, APP-1002, APP-1003, APP-1006, APP-1007, APP-1010 → APP-1001, APP-1002, APP-1003, APP-1004, APP-1006, APP-1007, APP-1010 why: new point-of-sale rollout in our Chicago office	Sep 27, 03:13:06 PM · prairie	done approved for the POS rollout

- 1

Ask for a change

Applications, the preferred hub or a second provider, with a reason.
- 2

Track it

Each request becomes a change request for the operators, and its progress is shown here.

Why it matters
Customers can request changes without email, and the request still goes through staff approval.

Self-service diagnostics

Prairie Grain Logistics your managed network service · site cust2

Your network Service levels Requests **Diagnostics**

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Diagnostics tests run now, from your own LAN host, to your applications and to our hubs

1 Test my applications 2 Test the hubs APP-1001 Unified Communications (VoIP) at hub-east Trace the path

Application	At	Answers	Round trip	Loss	HTTPS
APP-1001 Unified Communications (VoIP)	hub-east 192.168.71.10	yes	3.0 ms	0 %	not an HTTPS service
APP-1001 Unified Communications (VoIP)	hub-central 192.168.72.10	yes	3.4 ms	0 %	not an HTTPS service
APP-1001 Unified Communications (VoIP)	hub-west 192.168.73.10	yes	2.4 ms	0 %	not an HTTPS service
APP-1002 Email & Collaboration	hub-east 192.168.71.11	yes	2.7 ms	0 %	connects
APP-1002 Email & Collaboration	hub-west 192.168.73.11	yes	2.2 ms	0 %	connects
APP-1003 ERP — Finance & Procurement	hub-central 192.168.72.11	yes	2.0 ms	0 %	connects
APP-1004 Point of Sale Backend	hub-central 192.168.72.12	yes	1.8 ms	0 %	connects
APP-1004 Point of Sale Backend	hub-west 192.168.73.12	yes	1.1 ms	0 %	connects
APP-1006 Fleet Tracking & Telematics	hub-central 192.168.72.13	yes	1.1 ms	0 %	connects
APP-1007 Inventory & Warehouse Management	hub-central 192.168.72.14	yes	1.4 ms	0 %	connects
APP-1007 Inventory & Warehouse Management	hub-west 192.168.73.13	yes	1.4 ms	0 %	connects
APP-1010 Identity & Single Sign-On	hub-east 192.168.71.14	yes	2.5 ms	0 %	connects
APP-1010 Identity & Single Sign-On	hub-central 192.168.72.16	yes	1.7 ms	0 %	connects
APP-1010 Identity & Single Sign-On	hub-west 192.168.73.15	yes	1.0 ms	0 %	connects

5:51 PM · five pings to each, from your LAN host

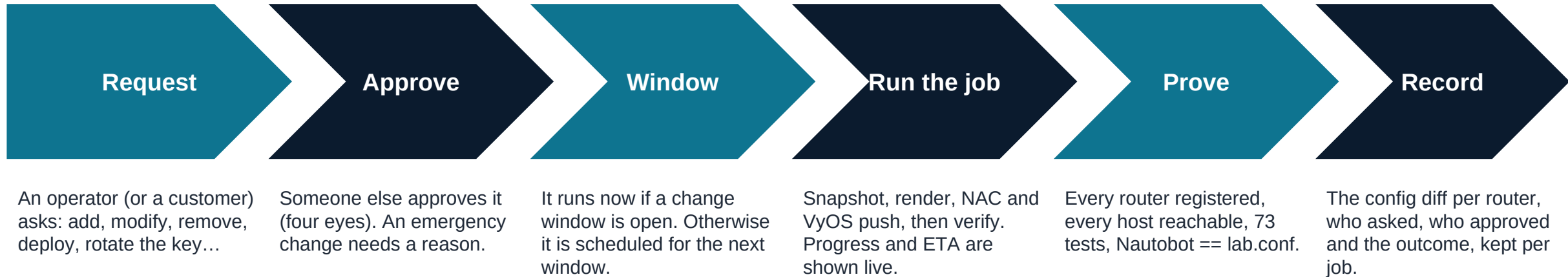
1 Tests from its LAN host
The customer tests its applications, the hubs, or traces a path.

2 Every application, every hub
Round-trip time, loss and HTTP status, run on demand.

Why it matters

The first-line diagnosis is done by the customer in seconds, with no ticket needed.

How a change moves through the portal



What is never allowed

- Approving your own request (four eyes)
- A viewer or customer starting a job
- Rotating the pre-shared key unless you are an admin
- A secret reaching git, a job log or the screen

What never needs approval

- Adding a customer: it takes nothing away
- Dry runs, tests and drift checks
- Customer diagnostics from its own LAN host
- Reading anything the role allows

Measured, not assumed

3 / 3

hubs every spoke is registered with

20 / 20

LAN hosts reach every other host

33 / 33

application checks passing

73 / 73

Robot tests pass after each change

0.6 s

worst failover when a hub fails; 29 of 35 flows never noticed

6.8 s

worst failover for dual-homed flows when a provider fails

54

IKE SAs re-authenticated with the new pre-shared key

5 m 28 s

key rotation end to end, one router at a time, then verified

Provider failure: dual-homed customers (cust1, cust4) move to cloud 2 over the second provider. Single-homed customers lose that provider by design, and the portal shows it as 'cut off' rather than hiding it.

Hub failure: customers keep two other hubs, live shortcuts need no hub, and routing converges when BGP gives up on the failed hub (hold 9 s).

In short

One source of truth

lab.conf drives the configuration, Nautobot and the portal, and drift is detected against it.

Every change is a job

It is planned, approved, run, verified and recorded with a before-and-after diff per router.

Resilience you can measure

Simulate a failure and get the outage timeline for every flow.

Customers see their own service

SLA, applications, requests and self-service diagnostics, and nothing else.

Secure by default

Roles, four eyes, key rotation on demand, and secrets never shown.